KASA HARENDRA

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Computer Science undergraduate (GPA 9.28) with experience in high-performance computing, GPU-accelerated machine learning, and full-stack AI applications. Proven ability to design robust, modular software systems in Python, Java, and C.

SKILLS AND ABILITIES

- **Languages:** Python, C, Java, MySQL
- **Technical Skills:** Web Scraping, Data Analysis using Python
- **Libraries/Frameworks:** PyTorch Pandas, Numpy, Scikit-Learn, Django, Langchain, FastAPI
- **Tools:** Git, Visual Studio Code, Jupyter Notebook
- **Platforms:** Linux, Windows
- **Soft Skills:** Problem-Solving

EXPERIENCE

Intern – High Performance Computing (HPC)
Centre for Development of Advanced Computing (CDAC), Pune
April 2025 – June 2025

- Successfully completed a 2-month internship focused on the sub-project **"Graph Representation and Visualization of Telecom Traffic Data."**
- Applied graph-based data structures and Python libraries (networkx) to represent and visualize complex telecom traffic datasets.
- Demonstrated inventiveness, strong problem-solving skills, and domain knowledge in HPC and data visualization.
- Gained hands-on experience with **GPU architecture and CUDA/OpenACC** for accelerating data visualization tasks.

PROJECTS

1. **High-Performance Multimodal RAG System | GenAI | Github**
   - **Tech Stack**: Python, Streamlit, LangChain, ChromaDB, Ollama API, ThreadPoolExecutor, DocumentLoaders.
   - **Implementation & Logic**: Built a sophisticated chat interface that enables context-aware interactions with diverse file types, including PDFs, images, spreadsheets, and Python scripts. The system features smart retrieval with automated source citations and conversation memory.
   - **Performance & Optimization**:
     - Achieved **3–10x faster processing** through concurrent embedding generation and batch processing using ThreadPoolExecutor.
     - Engineered a **parallel document ingestion pipeline** that processes multiple files simultaneously, resulting in a 2–5x speed improvement for large uploads.
     - Integrated **smart connection pooling** and system-aware worker configurations to minimize API latency and maximize resource utilization.

2. **Chat With Repo | Full-Stack AI | GitHub**
   - **Tech Stack:** FastAPI, React (Vite), Redis, ChromaDB, LangChain, Ollama, Docker Compose, Zustand, Tailwind CSS
   - **Implementation & Architecture:**
     - Engineered a **4-tier containerized application** with a novel **Tree-Based RAG system** that recursively traverses directory structures for highly precise code retrieval.
     - Designed a **fallback mechanism** to flat vector search *(k=3)* to ensure reliability when hierarchical retrieval is insufficient.
     - Built **stateful session management** using Redis with a **3600-second TTL**, implemented via custom FastAPI middleware.
   - **Performance & Optimization:**
     - Implemented a **session-aware caching layer** that instantly serves stored responses for repeated queries, eliminating redundant LLM computation and significantly reducing latency.
     - Secured the API using **IP-based rate limiting** *(30 requests/minute)* via slowapi.
     - Ensured **100% offline data privacy** by routing all inference tasks through a local Ollama LLM service.

3. **AI Data Tool (Conversational Tabular Analysis) | GenAI | Github**
   - **Tech Stack**: Python, Streamlit, Pandas, LangChain (PandasAgent), Matplotlib, Seaborn, Plotly, PythonREPL.
   - **Implementation & Logic**: Built a conversational platform that empowers users to analyze CSV and Excel datasets through plain English. The tool automates statistical summarization, data cleaning, and complex visualization generation.
   - **Performance & Optimization**:
     - Engineered a **transparent code execution** engine using PythonREPL, allowing for real-time validation and display of the generated Pandas logic.
     - Optimized prompt engineering for **zero-inference accuracy**, ensuring the agent strictly adheres to the data context.
     - Automated session-based **data cleanup** protocols to maintain system performance during high-intensity data processing.

4. **BERT Sentiment Analysis Engine | ML/DL | Github**
   - **Tech Stack**: Python, PyTorch, Hugging Face Transformers (BERT), Streamlit.
   - **Implementation & Logic**: Architected a binary classification model for sentiment analysis using a google-bert/bert-base-uncased backbone as a powerful feature extractor.
   - **Performance & Optimization**:
     - Enhanced model robustness with a **custom classification head** using **Dropout (0.25) regularization** to prevent overfitting.

. CERTIFICATIONS

- **CS50: Introduction to Programming with Python** – Harvard University
- **Generative AI Fundamentals** – Google Cloud Skill Boost
- **IBM Data Science Certificate** – IBM Cognitive Class
- **Competitive Programming –** NPTEL

Languages Known: English, Telugu, German(Intermediate)

EDUCATION

Kendriya Vidyalaya, Guntur — **Andhra Pradesh, India**
Class 10 (94.3%); Class 12 (90.4%); — 2013 - 2023

Vasireddy Venkatadri International Technological University — **Andhra Pradesh, India**
Bachelor of Technology (B.Tech); GPA: 9.43 — Aug 2023 - Ongoing